

Product Requirement Document (PRD)

Project Title: Real-Time Automated Seismic Wave Classification and Early Warning System

1. Overview

1.1 Purpose

This project aims to design and implement a real-time automated system capable of processing seismic waveform data in miniSEED (mseed) format, classifying seismic phases (P, S, and Surface waves), and estimating the epicenter and focus point of seismic events. The system further acts as an Earthquake Early Warning (EEW) platform that can provide alerts within seconds of event detection.

1.2 Background

Traditional earthquake detection systems (e.g., STA/LTA, threshold-based pickers) suffer from high false-alarm rates, latency, and difficulty in distinguishing seismic phases in noisy environments. Recent deep learning models such as EQTransformer, LEQNet, GL, and OBSTransformer have shown high accuracy in seismic event detection but lack full integration into real-time, low-latency systems. This PRD consolidates their strengths into a unified architecture designed for scalable, regional EEW deployment.

2. Problem Statement

Existing earthquake detection systems face several key limitations identified in literature: high false-positive rates due to noise and local disturbances, poor generalization to new geographic regions, inability to process waveforms in real-time on low-power devices, and lack of end-to-end integration between detection, classification, localization, and alerting. This project proposes a hybrid multi-stage ML system that overcomes these limitations while providing real-time seismic event classification and alerting capabilities.

3. Product Goals and Objectives

Goal	Objective	Success Metric
Real-time detection	Detect and classify seismic waves within 2 seconds of signal reception	Detection latency ≤ 2 s
Accuracy	Achieve $\geq 94\%$ precision and recall for P/S wave classification	F1 ≥ 0.94
Localization	Estimate epicenter within 15–20 km mean error	MAE ≤ 20 km
Noise robustness	Operate reliably under low SNR and teleseismic noise	False alarm rate < 2 /month
Scalability	Deployable on distributed network of seismic stations	< 1 MB model at edge
Early warning	Generate alerts automatically via APIs/SMS	End-to-end alert < 5 s

4. System Overview

4.1 Edge Layer

- Receives live waveform stream (miniSEED format) from local seismometer.
- Performs signal preprocessing: de-trending, de-meaning, bandpass filtering.
- Runs Lightweight CNN model (LEQNet-lite) for event detection and preliminary phase classification.
- Generates Pick Packets containing:
 - Timestamp
 - P/S phase probability
 - Short waveform snippet (10–30 s)
- Sends packets to the regional gateway via MQTT/gRPC

4.2 Regional Gateway Layer

- Aggregates Pick Packets from multiple edge stations.
- Uses ensemble of EQTransformer, TPhaseNet, and GL (Global–Local) models for refined phase picking and validation.
- Performs triangulation to compute:
 - Event epicenter (latitude, longitude)
 - Hypocenter depth (focus point)
 - Magnitude estimation (ML/empirical regression).
- Applies ensemble confidence scoring and teleseismic filtering to suppress false alerts.
- Issues regional alert messages upon event confirmation.

4.3 Cloud and Visualization Layer

- Stores waveform data and metadata in database (MongoDB).
- Runs periodic model training and retraining using new data (transfer learning).
- Provides real-time dashboards, maps, and public APIs for alert dissemination.
- Integrates SMS/email/webhook notification system for emergency alerts.

5. System Features

5.1 Core Functional Features

1. Waveform Ingestion: Continuous input from miniSEED files or seismic sensors.
2. Phase Classification: Classifies each segment into P, S, or Surface waves using deep learning.
3. Event Detection: Identifies seismic events above threshold confidence.
4. Triangulation: Calculates epicenter and focus point using P/S arrivals from multiple stations.
5. Alert Generation: Sends structured alert messages to clients and systems.

5.2 Supporting Features

- Noise Filtering: Adaptive denoising using learned filters and traditional bandpass.
- Self-Learning Module: Continuous model retraining using confirmed events.

- Data Storage: Efficient storage and retrieval of waveform data by event ID and station.
- Web Dashboard: Displays seismic map, event timeline, magnitude, and uncertainty.
- APIs: Provide third-party access for EEW applications and research use.

5.3 Advanced / Optional Features

- Single-station fallback: Predict distance and azimuth from one station using DL model (DisNet/AziNet concept).
- Shakemap Generation: Interpolates PGA/PGV for affected regions.
- Health Monitoring: Detects malfunctioning sensors or communication failures.
- Uncertainty Quantification: Outputs confidence interval for each prediction.
- Offline Mode: Temporarily store data when connection fails and sync later.

6. Functional Requirements

ID	Requirement Description	Priority
FR-1	System shall ingest continuous mseed waveform streams..	High
FR-2	System shall preprocess waveform data (de-mean, detrend, bandpass filter).	High
FR-3	System shall classify seismic data into P, S, and Surface waves in real time.	High
FR-4	System shall classify seismic data into P, S, and Surface waves in real time.	Medium
FR-5	System shall triangulate epicenter and focus point using multi-station data.	High
FR-6	System shall generate early warning alerts within 2 seconds.	High
FR-7	System shall store waveform and metadata in MongoDB.	High
FR-8	System shall visualize results on a dashboard (map + waveform).	Medium
FR-9	System shall quantify model uncertainty and attach confidence scores.	Medium
FR-10	System shall support auto-retraining with new confirmed events.	Low

7. Non-Functional Requirements

Category Requirement

Performance	End-to-end detection latency $\leq$ 2 seconds.
Reliability	99% uptime for detection nodes; message delivery acknowledgment.
Scalability	Support 100+ stations per region.
Security	Encrypted communication between nodes (TLS/SSL).
Portability	Deployable on ARM/Linux devices (Raspberry Pi, Jetson).
Maintainability	Modular microservice structure for updates and retraining.
Interoperability	Compatible with SEEDLink, ObsPy, and USGS data feeds.

8. System Components and Technologies

Component Technology Stack

Edge Detection	LEQNet-lite (TensorFlow Lite / ONNX)
Phase Picker	EQTransformer / TPhaseNet (PyTorch)
Communication	MQTT / gRPC / REST
Data Processing	ObsPy (Python library for seismology)
Database	MongoDB / InfluxDB
Visualization	React.js + Flask backend
Model Training	PyTorch / TensorFlow
Deployment	Docker / Kubernetes / Edge containers
Alert System	Twilio / Webhook / Email integration

9. Expected Outcomes

1. Signal Acquisition: miniSEED waveform received from seismometers.
2. Preprocessing: Filtering and normalization.
3. Edge Detection: LEQNet-lite detects potential events → sends Pick Packet.
4. Aggregation: Regional Gateway collects packets → EQTransformer refinement.

5. Triangulation: Uses P/S arrival times to compute epicenter and focus.
6. Classification Output: Saves separate mseed files for P, S, and Surface waves.
7. Alert Trigger: Gateway issues early warning (EEW message) if threshold met.
8. Visualization: Dashboard displays real-time event and shakemap.

10. Performance Metrics

Metric	Target Value
Detection Accuracy	$\geq 94\%$
False Alarm Rate	< 2 per month
P/S Arrival Time Error	< 0.5 s
Localization Error	< 20 km
Alert Latency	< 2 s
Edge Model Size	< 1 MB
Cloud Uptime	99%

13. Deliverables

1. Python-based prototype (v1.0) — mseed ingestion, classification, and epicenter estimation.
2. Gateway system (v2.0) — ensemble inference and alerting.
3. Web dashboard (v3.0) — live visualization and alert management.
4. Deployment package (v4.0) — containerized deployment for testing.

14. Expected Outcomes

- A fully functional prototype capable of classifying P, S, and Surface waves in real time.
- A scalable EEW architecture deployable at regional or national level.
- Demonstrated improvement in accuracy, robustness, and alert latency compared to traditional and single-model systems.